HARSH SHARMA

ENTRY-LEVEL AI/ML & BACKEND ENGINEER

Vadodara, Gujarat | +91 6352532739 | harshsharmag99@gmail.com
LinkedIn: linkedin.com/in/harshxo44 | GitHub: github.com/Harshxo44 | Portfolio: codewithaesthetic.vercel.app

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science and Engineering student (Parul University, 2027) with hands-on experience building AI-enabled media pipelines and secure backend services. Proficient in Python speech-processing tools, Java/Spring Boot and Node.js/Express REST APIs, asynchronous job handling, and cloud-GPU experimentation. Seeking graduate or internship roles in AI application development, AI/ML engineering, or backend software engineering.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript (ES6+), Kotlin, C++, Dart
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Vite
Backend & APIs: Spring Boot, Spring Security, Spring Data JPA, JWT, Node.js, Express.js, Flask, REST APIs
AI / Speech Processing: Faster Whisper, XTTS v2, Demucs, librosa, FFmpeg, Rubber Band, Edge TTS, gTTS
AI Workflow Integration: Speech-to-text, text translation, text-to-speech, voice-cloning model integration, vocal separation, audio mixing, prosody processing, intent routing, tool dispatch, response validation
Cloud & Development Tools: Google Colab GPU (T4 workflow), ngrok, Git, GitHub, Maven, JUnit, Postman
Databases & App Platforms: MySQL, PostgreSQL, H2, MongoDB, Firebase Authentication, Cloud Firestore, Flutter
Embedded & Connectivity: ESP32 firmware (C++), BLE GATT, OBD-II
Cloud / Certifications: AWS Cloud Practitioner; IBM and Microsoft AI/data learning credentials

PROJECTS

Velora — Connected Vehicle Intelligence Platform (Prototype) | Java 21, Spring Boot, Spring Security, JWT, PostgreSQL, Kotlin, ESP32, BLE
GitHub: github.com/Harshxo44/Velora

- Developed a Spring Boot REST backend with JWT authentication covering vehicle registration, device pairing, trip tracking, telemetry ingestion, diagnostic trouble code lookup, maintenance reminders, and privacy and data-purge endpoints.
- Designed a deterministic AI orchestration layer with intent routing, tool dispatch, and grounded response synthesis, including pre-response safety checks for critical coolant-temperature and battery-voltage conditions.
- Built integration layers for an Android BLE/Retrofit client and ESP32 firmware modules (BLE GATT, OBD-II PID parsing, telemetry encoding) within a read-only design that issues no vehicle control commands.
- Configured H2 for local development and PostgreSQL for deployment; planned extensions include automotive RAG (FAISS) and telemetry-based predictive maintenance.

DubAI Studio — AI Video Dubbing Platform | Python, Node.js, Express, FFmpeg, Faster Whisper, XTTS v2, Demucs
GitHub: github.com/Harshxo44/DubAI_Studio

- Engineered an end-to-end AI video-dubbing pipeline that extracts audio, separates vocals, transcribes and translates speech, synthesizes dubbed audio, and remixes it with the original video.
- Integrated Python speech-processing components, including Faster Whisper, XTTS v2, Demucs, librosa-based prosody processing, and FFmpeg utilities.
- Implemented a Node.js/Express backend for uploads and asynchronous job status and results, with a Python bridge supporting local CPU execution and a Google Colab GPU workflow; documented support for 11 language options.

EXPERIENCE

AI/ML Intern (Virtual) — CodSoft | July 2026

- Developed a Random Forest model to predict bank customer churn using the Kaggle Churn Modelling dataset.
- Built a classifier to detect fraudulent credit card transactions using the Kaggle fraud-detection dataset with separate training and test files.
- Classified movie genres from plot summaries using TF-IDF features and Logistic Regression on the IMDb genre dataset.
- Structured each project as a Jupyter/Colab notebook plus a standalone Python script with saved model files and a requirements.txt for reproducible execution.

EDUCATION

B.Tech, Computer Science and Engineering | Parul University, Vadodara | 2023–2027
Relevant coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services
- IBM Java Developer Professional Certificate
- IBM AI Engineering Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Microsoft AI & ML Engineering Professional Certificate